

Self-supervised reproductive life-course phenotyping in the National Survey of Family Growth: a temporal public-use survey study

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Abstract

Background: Reproductive health across adolescence and adulthood is shaped by pregnancy history, contraceptive behavior, partnership context, fertility care, impaired fecundity, insurance, and socioeconomic conditions. Public National Survey of Family Growth (NSFG) analyses often focus on single domains, leaving multivariable reproductive life-course heterogeneity less directly characterized.

Methods: We harmonized public-use NSFG female respondent and female pregnancy files from 2011–2013, 2013–2015, 2015–2017, 2017–2019, and 2022–2023. The primary analysis included females aged 15–44 years. Female pregnancy files were aggregated to respondent-level histories and linked by CaseID. A leakage-controlled masked tabular self-supervised learning (SSL) encoder was trained for 30 epochs on 2011–2017 records, phenotype selection was performed in 2017–2019, and 2022–2023 was reserved for within-NSFG temporal validation rather than validation in a separate health-system dataset. Endpoint-enrichment intervals used a stratified cluster bootstrap based on public-use VEST/VECL design variables.

Results: The harmonized matrix included 26,478 respondents and 598 cross-cycle columns before feature selection. The training/pretraining, development, and temporal-validation splits included 16,176, 5,409, and 4,893 respondents, respectively. The primary encoder used 48 endpoint-excluded features and generated 48-dimensional respondent embeddings. Development-cycle clustering selected three phenotypes (silhouette 0.515; Davies–Bouldin 0.795; bootstrap adjusted Rand index 0.996). In 2022–2023, P0 was a small phenotype with higher survey endpoint burden, enriched for adverse pregnancy-history proxy (prevalence ratio [PR] 4.11; 95% CI 2.96–5.47), mistimed or unwanted pregnancy history (PR 2.23; 95% CI 2.05–2.43), fertility or pregnancy-loss help (PR 1.68; 95% CI 1.20–2.21), and impaired fecundity (PR 1.54; 95% CI 1.31–1.78). P1 showed the highest enrichment for contraceptive vulnerability (PR 1.33; 95% CI 1.22–1.45), whereas P2 represented a younger low-pregnancy-exposure phenotype.

Conclusions: A masked tabular SSL workflow can convert public-use NSFG respondent and pregnancy files into interpretable reproductive life-course phenotypes. The results support representation learning, phenotype discovery, and endpoint enrichment as reproducible tools for population reproductive-health surveys; they do not establish clinical diagnosis, individual treatment guidance, or causal mechanisms.

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1 Introduction

Reproductive health is rarely determined by a single exposure. Contraceptive use, pregnancy timing, pregnancy outcomes, partnership history, fertility services, impaired fecundity, health insurance, and socioeconomic conditions can co-occur across the reproductive life course. Analyses that isolate one domain are valuable for surveillance, but they may understate heterogeneity among respondents whose reproductive histories and social conditions jointly shape reproductive-health needs.

The National Survey of Family Growth (NSFG) is a nationally representative United States survey that gathers information on fertility, contraception, pregnancy and births, marriage and cohabitation, infertility, and reproductive health. The public-use female respondent file can be linked to the female pregnancy file, enabling respondent-level summaries of pregnancy histories while preserving a reproducible public-data workflow. Recent NCHS reports using the 2022–2023 NSFG have described current contraceptive status, fertility-service use, and birth expectations. These reports provide essential national estimates, but they are not designed to learn multivariable reproductive life-course phenotypes across respondent and pregnancy records.

Self-supervised learning for tabular data offers a practical approach for learning representations from sparse, heterogeneous survey variables without using a single supervised label during representation training. Prior tabular SSL and transformer methods, including VIME, TabTransformer, SAINT, and SCARF, support the use of masking, contextual feature embeddings, and corruption-based pretraining for tabular representation learning. We adapted this idea to the public-use NSFG setting with explicit temporal splitting and endpoint leakage control.

This study aimed to construct a respondent-level reproductive life-course matrix from public NSFG files, train a masked tabular SSL encoder using earlier cycles, identify interpretable reproductive phenotypes in a development cycle, and evaluate phenotype-associated enrichment of prespecified reproductive-health endpoints in the temporally held-out 2022–2023 release.

2 Methods

2.1 Data sources and study population

We used public-use NSFG female respondent and female pregnancy files from 2011–2013, 2013–2015, 2015–2017, 2017–2019, and 2022–2023. Fixed-width 2011–2019 files were parsed with official CDC Stata dictionaries. The 2022–2023 public-use CSV files and user guide were obtained from the CDC/NCHS NSFG public-use data page. The primary analysis was restricted to females aged 15–44 years to preserve comparability between 2011–2019 and 2022–2023 public-use releases.

2.2 Temporal split and leakage control

The 2011–2013, 2013–2015, and 2015–2017 cycles were used for SSL pretraining. The 2017–2019 cycle was used for phenotype model selection and development-cycle diagnostics. The 2022–2023 cycle was reserved for within-NSFG temporal validation, and its endpoint labels were used only for final evaluation. This design evaluates temporal transport within a public-use survey series, not validation in a separate health-system cohort or registry. Survey weights were retained for descriptive prevalence and phenotype profiles, but were not used as model inputs. For the primary encoder, variables directly defining the validation endpoints were excluded before feature selection to reduce endpoint leakage.

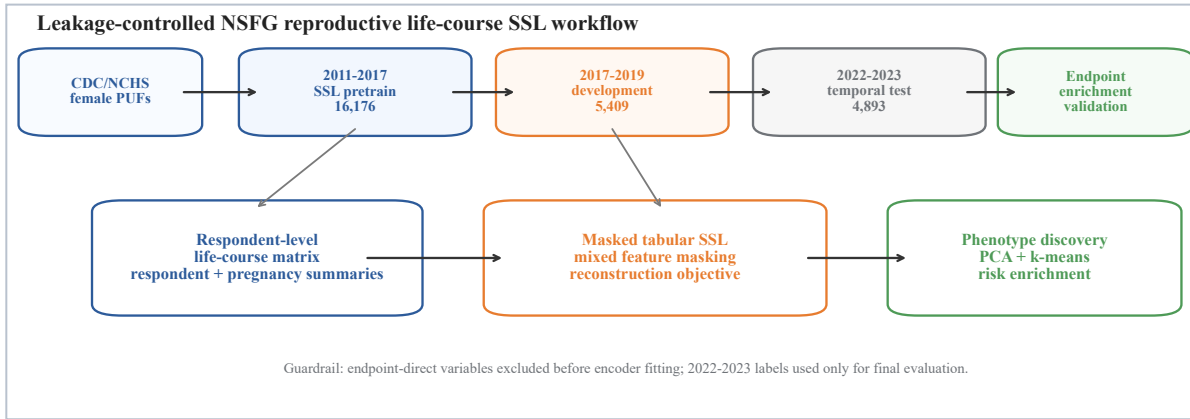


Figure 1: Study design and leakage-controlled analysis workflow. Public-use NSFG female respondent and female pregnancy files were harmonized across cycles. Earlier cycles were used for masked SSL pretraining, 2017–2019 for phenotype selection, and 2022–2023 for within-NSFG temporal validation. Endpoint-direct variables were excluded from the primary encoder input.

2.3 Respondent-level life-course matrix

Female pregnancy records were aggregated to respondent level using pregnancy counts, live-birth and pregnancy-outcome counts, pregnancy-intention summaries, low-birthweight or stillbirth proxies, and prenatal-care or smoking indicators where public-use variables were available. These summaries were joined to female respondent variables by CaseID. Candidate input domains included demographics and social factors, partnership and marriage, reproductive timing, sexual and contraceptive history, pregnancy-history summaries, fertility and reproductive-health indicators, insurance and poverty, and missingness or skip-pattern indicators.

Table 1: Input feature set and validation endpoint definitions. Direct endpoint-defining variables were excluded from the primary encoder input before SSL fitting.

Domain or endpoint	Operational definition	Features	Leakage-control rule
Primary SSL input features	Endpoint-direct variables excluded; top nonconstant features selected by training-cycle completeness and variance.	48	Direct endpoint regex excluded before encoder fitting.
Contraceptive vulnerability	CONSTAT1 code 42; denominator is the NSFG analytic female respondent population, interpreted as a current vulnerability proxy rather than an incident pregnancy outcome		Direct endpoint-variable regex excluded; full regex retained in source-data table.
Fertility or loss help	HLPPRG code 1 or HLPMP code 1, indicating ever receiving medical help to become pregnant or to prevent miscarriage		Direct endpoint-variable regex excluded; full regex retained in source-data table.
Mistimed/unwanted pregnancy history	Among respondents with pregnancy records, any NEWWANTR code 3/4/6, WANTRESP code 3/5, WANTEDPART code 3/5, TIMINGOK code 3, or WANTBOLD code 5		Direct endpoint-variable regex excluded; full regex retained in source-data table.
Adverse pregnancy-history proxy	Among respondents with pregnancy records, any public-use low-birthweight live birth indicator (LBW1 code 1) or stillbirth outcome (OUTCOME code 3)		Direct endpoint-variable regex excluded; full regex retained in source-data table.
Impaired fecundity	FECUND code 1-5 or INFERT code 1-2, using public-use NSFG fecundity or infertility recodes		Direct endpoint-variable regex excluded; full regex retained in source-data table.

2.4 Masked tabular SSL encoder

The primary model used a masked tabular SSL encoder implemented for numeric-coded public-use survey variables. Feature values were median-imputed and standardized using training-cycle parameters, and missingness indicators were supplied as token inputs. During training, 15% of observed values were masked and the model minimized masked reconstruction mean squared error. The compact encoder used 1 transformer encoder layer, hidden dimension 32, 4 attention heads, dropout 0.10, batch size 1024, AdamW optimization with learning rate 0.001, and 30 training epochs. The final analysis artifacts used 48 endpoint-excluded input features and 48-dimensional respondent embeddings. This model was designed as a compact public-survey representation learner, not as a large foundation model.

2.5 Phenotype discovery and validation

Respondent embeddings were projected to principal components fitted on training-cycle embeddings. The number of phenotypes was selected in 2017–2019 by comparing $k=2-8$ using silhouette score, Davies–Bouldin index, minimum cluster proportion, and 60 bootstrap adjusted Rand index (ARI) resamples. Development-cycle centroids were then fixed, and 2022–2023 respondents were assigned to the nearest centroid. Phenotype descriptions were based on survey-weighted profile characteristics, not on outcome enrichment alone.

Five validation endpoints were prespecified: contraceptive vulnerability, fertility-service or pregnancy-loss help, mistimed or unwanted pregnancy history, adverse pregnancy-history proxy, and impaired fecundity status. The contraceptive-vulnerability endpoint was defined from the NSFG current contraceptive-status recode and interpreted as a current vulnerability proxy within the analytic respondent population rather than an incident pregnancy outcome. Pregnancy-history endpoints used respondent-level pregnancy-file summaries and were interpreted among respondents with available pregnancy-history information. Endpoint enrichment was summarized using survey-weighted prevalence ratios and risk differences. Uncertainty intervals for endpoint enrichment were estimated with a stratified cluster percentile bootstrap using public-use VEST strata and VECL cluster variables, retaining the public-use respondent weights in each resample.

Secondary supervised endpoint-enrichment models compared phenotype-only, SSL-embedding, and SSL-plus-phenotype inputs using L2-regularized logistic regression. Models were trained on 2011–2019 records, features were standardized within the training data, class weights were balanced, and 2022–2023 labels were used only for final evaluation. AUPRC, AUPRC enrichment over baseline prevalence, and AUROC were reported as unweighted model-performance summaries, whereas phenotype prevalence profiles and endpoint-enrichment estimates used survey weights. These supervised summaries were used to assess whether learned representations enriched survey endpoints; they were not interpreted as bedside diagnostic models, individual prognostic models, treatment-decision tools, or calibrated absolute-risk tools.

2.6 Robustness and sensitivity analyses

Five prespecified robustness analyses were added after the primary workflow was fixed. First, because the 2022–2023 NSFG release includes females aged 15–49 years whereas earlier public-use cycles were restricted to ages 15–44 years, we transferred the trained endpoint-excluded encoder to the full 2022–2023 age range and compared endpoint enrichment in ages 15–44 versus 15–49 years without refitting the encoder, PCA, or development-cycle centroids. Second, we fitted survey-weighted one-vs-rest logistic enrichment models for each endpoint and phenotype, adjusting for age, race/ethnicity, education, poverty, insurance, and parity; confidence intervals used a stratified

cluster bootstrap with VEST/VECL. These adjusted models were descriptive enrichment checks rather than causal models. Third, we compared SSL phenotyping with raw matrix PCA plus k-means, MCA-style one-hot SVD plus k-means, and selected-variable Bernoulli latent class analysis. Fourth, we trained a full-domain SSL sensitivity encoder that did not exclude endpoint-direct variables and compared enrichment with the primary endpoint-excluded encoder. Fifth, subgroup robustness was summarized across age group, race/ethnicity, poverty, insurance, and parity strata.

3 Results

3.1 Cohort and harmonized matrix

The harmonized analysis included 26,478 respondents across five public-use cycles (Table 2). The temporal-validation cohort included 4,893 respondents from 2022–2023. Pregnancy records were linked by CaseID in each cycle, and the harmonized matrix contained 598 columns before primary feature selection (Figure 2).

Table 2: Cohort characteristics by temporal analysis split. Weighted estimates use public-use NSFG analysis weights; model training did not use survey weights as input features.

Split	N	Age	Parity	Pregnancy record, %	Contraceptive vulnerability, %	Fertility/loss help, %	Mistimed/unwanted pregnancy, %	Adverse pregnancy proxy, %	Impaired fecundity, %
Training/pretraining	16,176	29.6	1.19	61.1	7.2	11.5	47.8	9.9	32.3
Development/model selection	5,409	29.6	1.09	56.4	7.2	10.4	44.7	8.5	32.6
Temporal validation	4,893	29.6	0.99	50.0	7.8	11.1	34.2	5.4	30.4

3.2 SSL phenotype discovery

The masked reconstruction loss declined across 30 SSL training epochs, supporting optimization of the compact encoder while remaining a training diagnostic rather than an outcome-validation metric (Figure 7). In the development cycle, k=3 had the highest silhouette score (0.515) and high bootstrap ARI (0.996) among candidate cluster counts. The smallest development-cycle cluster comprised 4.8% of respondents, below the preferred 5% threshold; therefore, k=3 was treated as a development-selected, bootstrap-stable, small-cluster solution rather than evidence of a universal reproductive taxonomy (Figure 3; Supplementary Table 5).

3.3 Temporal-validation phenotype profiles

In 2022–2023, P0 was a small phenotype with older age, high parity, extensive pregnancy history, and higher survey endpoint burden, including the highest adverse pregnancy-history proxy and impaired fecundity prevalence. Its weighted mean age was 36.1 years and parity was 2.54; the adverse pregnancy-history proxy and mistimed or unwanted pregnancy-history prevalence were 22.1% and 76.2%, respectively. P1 was also pregnancy-exposed but had a larger cluster size and the highest contraceptive vulnerability prevalence. P2 was younger and low pregnancy-exposure, with weighted mean age 24.8 years and parity 0.04. These profile-based patterns are shown in Figure 4 and Table 3.

3.4 Endpoint enrichment and secondary supervised validation

Phenotype enrichment was strongest for P0 across pregnancy-history and fecundity endpoints. P0 had PR 4.11 for adverse pregnancy-history proxy (bootstrap 95% CI 2.96–5.47), PR 2.23 for mistimed or unwanted pregnancy history (95% CI 2.05–2.43), PR 1.68 for fertility-service or pregnancy-loss help (95% CI 1.20–2.21), and PR 1.54 for impaired fecundity (95% CI 1.31–1.78). P1 showed

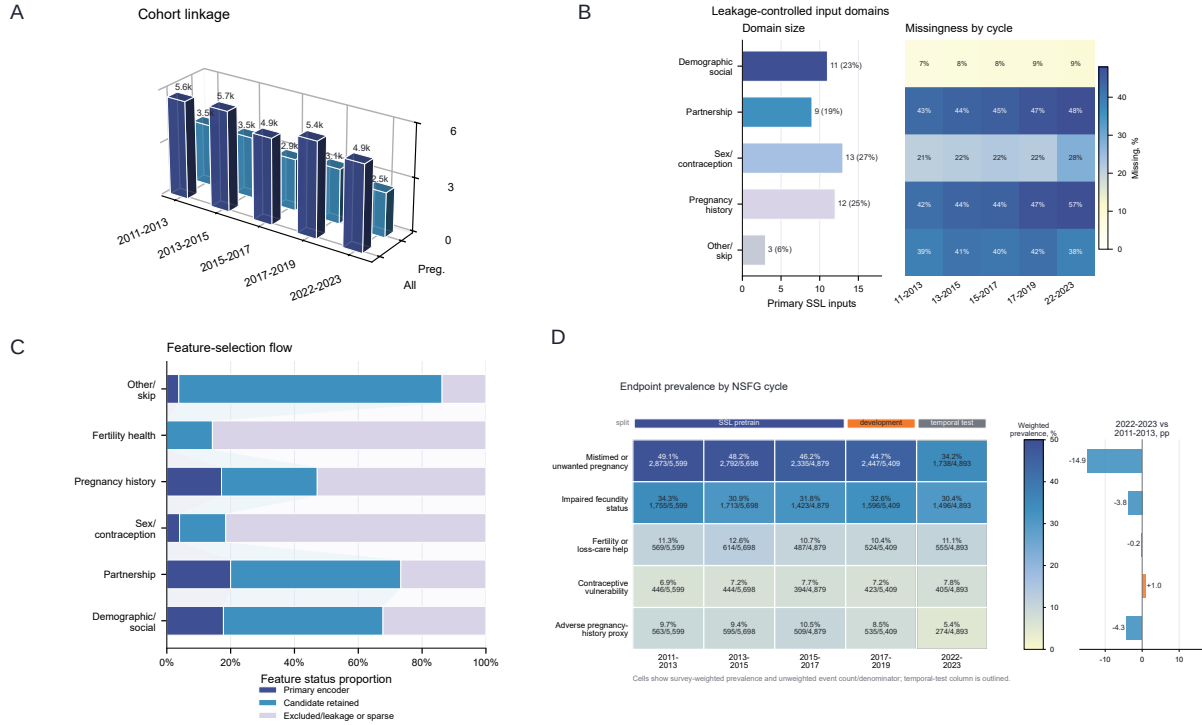


Figure 2: Harmonized cohort structure, leakage-controlled inputs, feature selection, and endpoint prevalence. (A) Female respondent and pregnancy-record linkage counts across NSFG cycles. (B) Primary SSL input-domain sizes and domain-level missingness by cycle. (C) Feature-selection flow separating primary encoder inputs, retained candidate features, and excluded or sparse variables. (D) Weighted endpoint prevalence across cycles with temporal-test change relative to 2011–2013.

Table 3: Survey-weighted temporal-validation phenotype profiles in 2022–2023. Phenotype labels were assigned from development-cycle centroids and named descriptively from profile characteristics, not from outcome status alone.

Profile characteristic	P0	P1	P2
Age, years	36.14	34.42	24.78
Parity	2.54	1.94	0.04
Pregnancy records	3.49	2.61	0.05
Any pregnancy record, %	95.4	100.0	3.2
Poverty-income ratio	230.11	278.87	340.28
Contraceptive vulnerability, %	7.4	10.4	5.6
Fertility or loss help, %	18.7	18.2	4.3
Mistimed/unwanted pregnancy, %	76.2	67.0	2.3
Adverse pregnancy-history proxy, %	22.1	9.2	0.5
Impaired fecundity, %	47.0	42.7	18.4

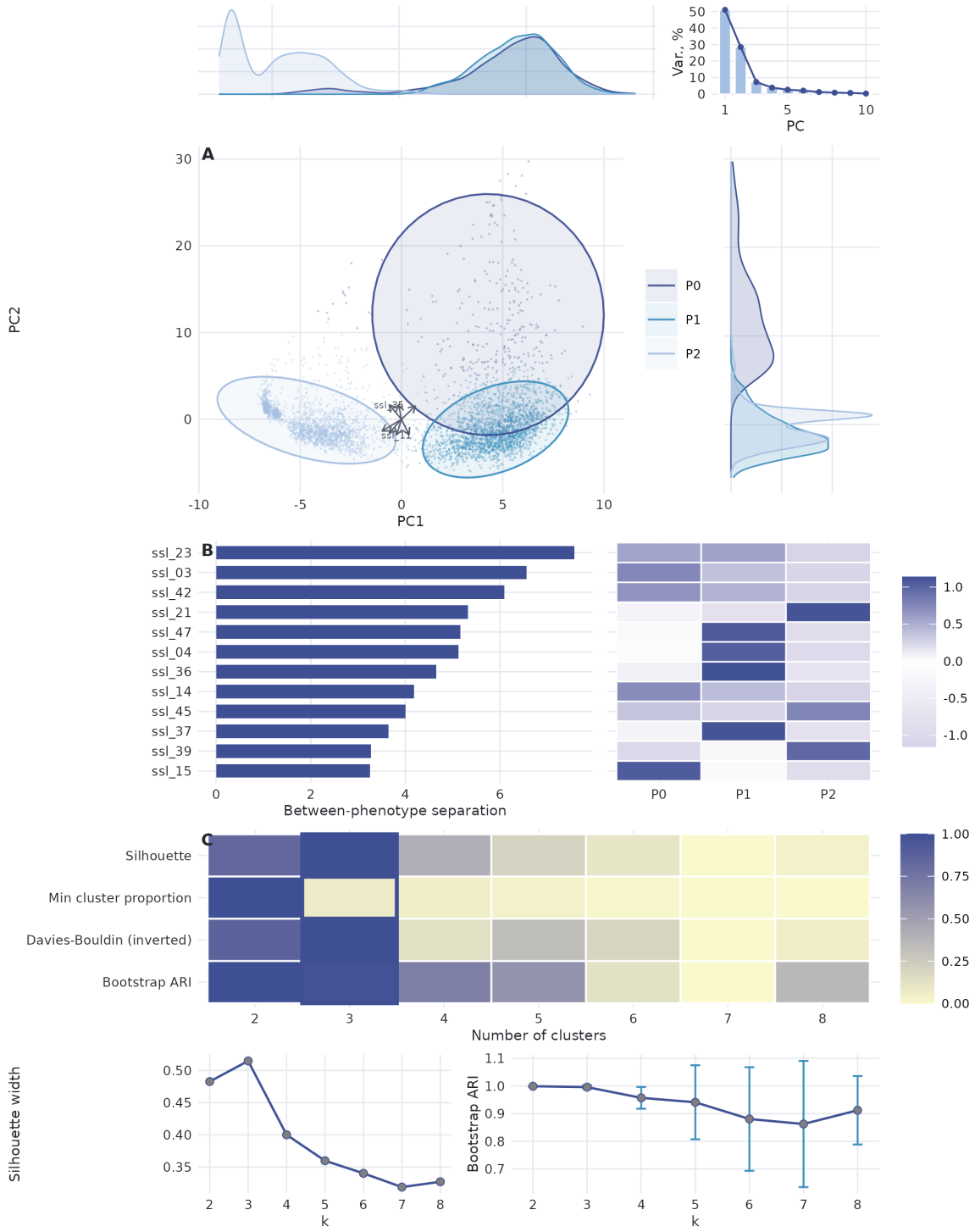


Figure 3: SSL embedding and phenotype discovery. (A) PCA visualization of 2022–2023 SSL embeddings colored by transferred phenotype, with phenotype ellipses, marginal PC density, leading embedding loadings, and a scree summary. (B) Leading embedding drivers and phenotype-level mean patterns. (C) Development-cycle multi-metric heatmap for cluster-number selection. (D) Silhouette width and bootstrap adjusted Rand index across candidate cluster numbers.

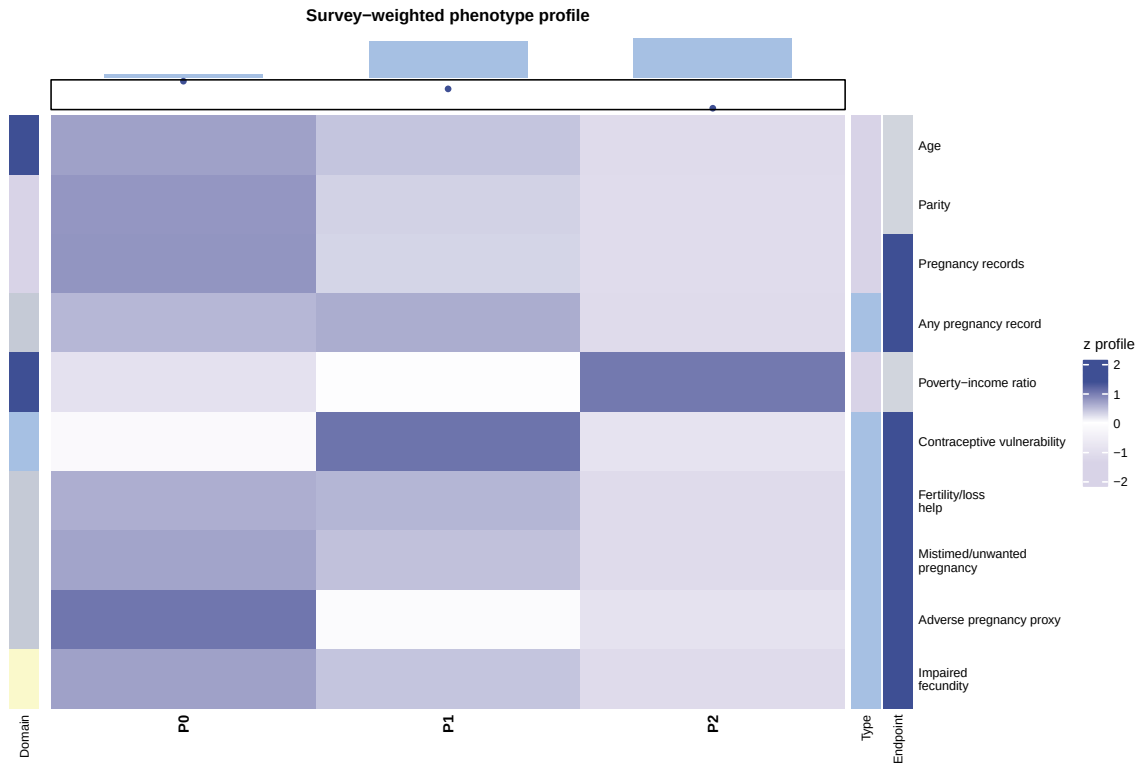


Figure 4: Survey-weighted phenotype profiles in 2022–2023. Heatmap cells show standardized deviations across phenotypes for selected reproductive life-course descriptors and endpoint-related characteristics; cell labels show weighted raw means or prevalences. Top and side annotation bars summarize phenotype burden and variable-domain grouping.

the highest enrichment for contraceptive vulnerability (PR 1.33, 95% CI 1.22–1.45), consistent with its larger pregnancy-exposed profile and higher current contraceptive vulnerability prevalence. P2 was depleted across pregnancy-history endpoints because it represented a younger, low-pregnancy-exposure phenotype.

Secondary supervised summaries showed that SSL embeddings, rather than phenotype labels alone, carried most of the endpoint-enrichment information for several endpoints. Adding phenotype labels to SSL embeddings did not consistently improve AUPRC enrichment, with the strongest embedding-based enrichment observed for contraceptive vulnerability, adverse pregnancy-history proxy, and mistimed or unwanted pregnancy history (Figure 5; Table 4). These results support the value of learned survey representations for population-level endpoint enrichment and phenotype discovery, but they should not be interpreted as clinical diagnostic accuracy or superiority over all supervised modeling strategies.

Table 4: Temporal-validation endpoint enrichment and secondary supervised model metrics. The highest-enrichment phenotype is shown for each endpoint. Prevalence-ratio intervals are stratified cluster bootstrap percentile intervals using public-use VEST/VECL design variables with respondent weights retained. Supervised models are interpreted as endpoint-enrichment summaries rather than diagnostic models; the best feature set denotes the highest point-estimate AUPRC enrichment.

Endpoint	Top phenotype	Top prevalence, %	Baseline, %	PR (95% CI)	RD, %	Best AUPRC set	AUPRC enrichment	AUROC
Adverse pregnancy-history proxy	P0	22.1	5.4	4.11 (2.96–5.47)	16.7	SSL + phenotype	3.43	0.83
Contraceptive vulnerability	P1	10.4	7.8	1.33 (1.22–1.45)	2.6	SSL embedding	5.71	0.90
Fertility or loss help	P0	18.7	11.1	1.68 (1.20–2.21)	7.6	SSL + phenotype	2.44	0.78
Impaired fecundity	P0	47.0	30.4	1.54 (1.31–1.78)	16.6	SSL + phenotype	1.78	0.74
Mistimed/unwanted pregnancy history	P0	76.2	34.2	2.23 (2.05–2.43)	42.0	SSL + phenotype	2.34	0.92

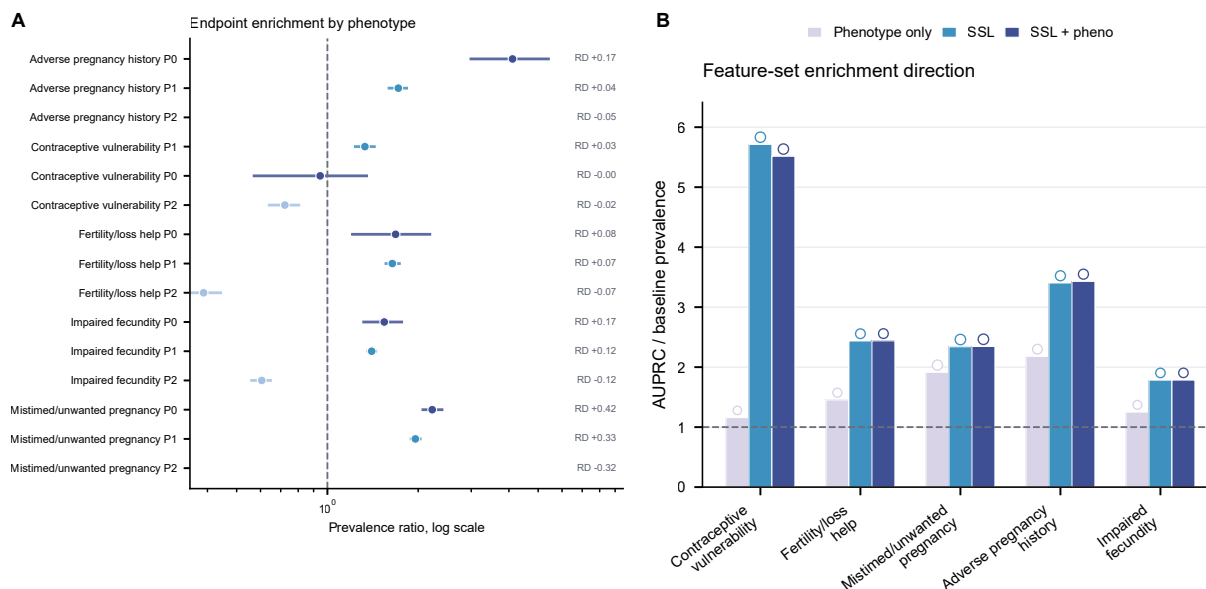


Figure 5: Endpoint enrichment and secondary supervised endpoint-enrichment summaries. (A) Survey-weighted prevalence ratios by phenotype in the 2022–2023 within-NSFG temporal-validation cohort with stratified cluster bootstrap confidence intervals. The dashed vertical line indicates no enrichment. (B) AUPRC enrichment over baseline prevalence for phenotype-only, SSL-embedding, and SSL-plus-phenotype summaries.

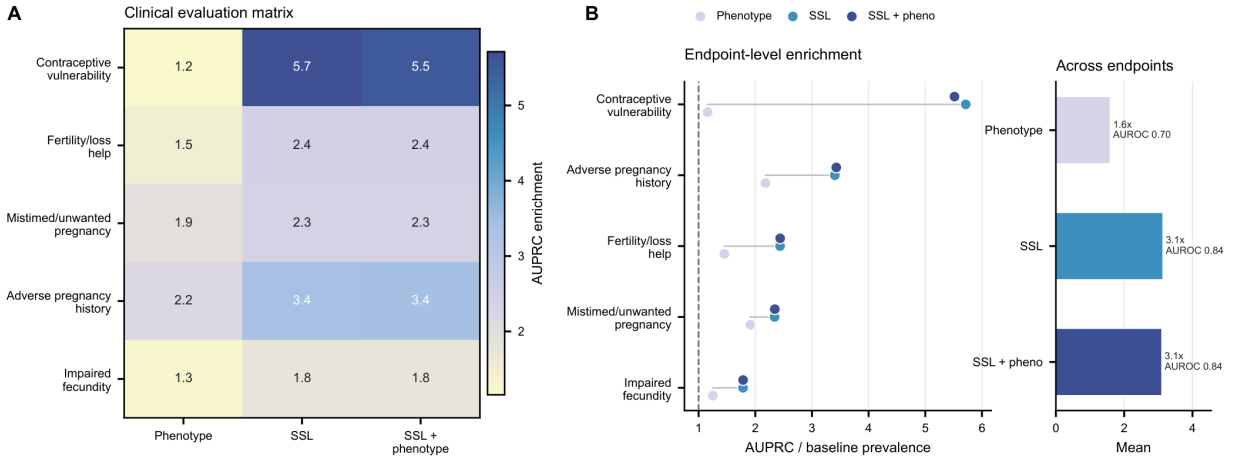


Figure 6: Secondary model diagnostics. (A) AUROC matrix for phenotype-only, SSL-embedding, and SSL-plus-phenotype summaries across validation endpoints. AUROC is reported as a secondary discrimination summary and is not the primary claim. (B) Endpoint-level and across-endpoint AUPRC enrichment summaries, highlighting that the SSL representation supports survey endpoint enrichment and phenotyping rather than diagnostic prediction.

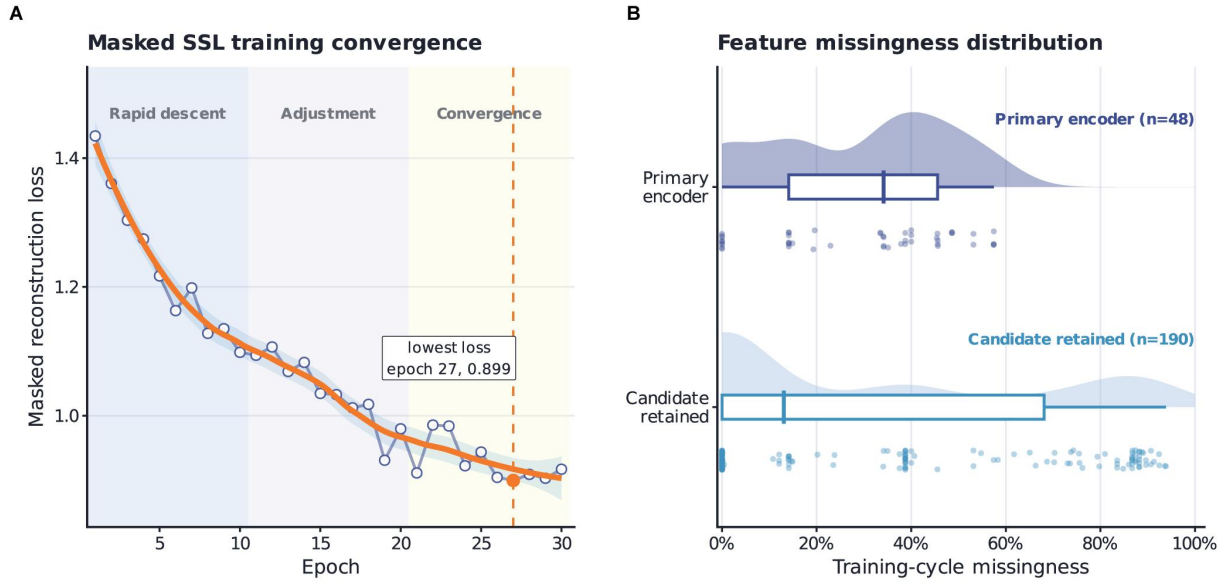


Figure 7: SSL diagnostics. (A) Masked reconstruction loss across SSL training epochs, with rapid-descent, adjustment, and convergence phases, a LOESS-smoothed trend, smoothing band, and minimum-loss epoch annotation. (B) Raincloud distribution of training-cycle missingness among primary encoder features and retained candidate SSL features. The smoothing band is a visualization aid and is not an external validation interval.

3.5 Robustness and sensitivity analyses

Age-range sensitivity analyses preserved the main enrichment direction after including 45–49-year-old respondents from the 2022–2023 public-use release. The reassigned 15–44 subset exactly reproduced the primary temporal-validation phenotype labels (ARI 1.00). The highest phenotype PR for adverse pregnancy-history proxy attenuated from 4.11 in ages 15–44 to 3.25 in ages 15–49, and the highest PR for mistimed or unwanted pregnancy history changed from 2.23 to 1.98; all prespecified endpoints retained the same enrichment direction in the expanded age range (Supplementary Table 7; Supplementary Figure 8).

Covariate-adjusted enrichment models showed that phenotype associations were not solely explained by age, race/ethnicity, education, poverty, insurance, or parity. P0 remained enriched for adverse pregnancy-history proxy after adjustment (adjusted OR 2.05, 95% CI 1.21–3.63). P1 remained enriched for contraceptive vulnerability (adjusted OR 1.85, 95% CI 1.40–2.44), fertility-service or pregnancy-loss help (adjusted OR 2.50, 95% CI 1.91–3.55), and mistimed or unwanted pregnancy history (adjusted OR 9.64, 95% CI 7.72–12.92). These adjusted models support descriptive robustness but do not imply causal phenotype effects (Supplementary Table 8).

Baseline method comparisons supported the stability of the SSL phenotype representation. SSL embedding plus k-means had development bootstrap ARI 0.953 and a minimum temporal-validation cluster proportion of 4.9%. Raw matrix PCA plus k-means produced a much smaller minimum cluster proportion (0.1%), indicating that some high enrichment values from raw PCA reflected unstable tiny clusters. Selected-variable Bernoulli LCA had development bootstrap ARI 0.904 but lower average endpoint-enrichment summaries than SSL. In leakage sensitivity, the full-domain encoder did not uniformly exceed the endpoint-excluded encoder; for adverse pregnancy-history proxy, the maximum PR was 4.11 for the primary endpoint-excluded encoder versus 2.01 for the full-domain encoder. Subgroup summaries showed enrichment patterns across demographic and reproductive-history strata, with some extreme subgroup values arising from smaller cells and therefore treated as descriptive robustness evidence (Supplementary Tables 9–11; Supplementary Figures 9–10).

4 Discussion

This temporal public-use survey study shows that masked tabular SSL can transform NSFG female respondent and pregnancy files into interpretable reproductive life-course phenotypes. The main contribution is not a new single-endpoint risk score or clinical prediction model, but a reproducible workflow for learning survey-based respondent representations and translating them into profile-based phenotypes with within-NSFG temporal validation.

The three-phenotype solution separated a small group with higher survey endpoint burden, a larger pregnancy-exposed group with contraceptive vulnerability enrichment, and a younger low-pregnancy-exposure group. This structure is consistent with the descriptive reproductive-health logic that survey-measured reproductive needs emerge from combinations of age, pregnancy history, partnership and contraceptive context, fertility services, and impaired fecundity rather than from any single variable. The phenotype interpretation remained profile-based, avoiding naming clusters solely by their validation endpoints.

The study differs from prior NSFG reports and analyses that focus on specific domains such as contraceptive status, fertility services, birth expectations, adolescent pregnancy intention classes, IUD use across life stages, or contraceptive non-use. Those analyses remain essential for domain-specific surveillance and policy; our analysis adds a multivariable representation-learning layer that can summarize cross-domain reproductive life-course heterogeneity in public-use survey data.

Several limitations are important. First, NSFG public-use data are self-reported and do not contain hospital-verified clinical outcomes. Endpoint definitions therefore rely on public-use survey recodes and should be interpreted as reproductive-health proxies. Second, validation was temporal within the NSFG public-use survey series rather than external validation in an independent survey, hospital EHR, or registry. Third, the higher-burden P0 phenotype was small in the development and temporal-validation cohorts; this supports interpretability but limits precision and motivates replication. Fourth, endpoint leakage was reduced by direct-variable exclusion and was examined with a full-domain sensitivity encoder, but public-use survey variables are correlated and indirect leakage cannot be fully eliminated. Fifth, adjusted enrichment and subgroup analyses reduce concern that phenotypes merely restate age or parity gradients, but they remain descriptive robustness checks and do not identify causal mechanisms. Finally, the compact encoder was designed for reproducible CPU-feasible public-data analysis and should not be framed as a large foundation model.

5 Conclusions

A leakage-controlled masked tabular SSL workflow identified interpretable reproductive life-course phenotypes in public-use NSFG data and demonstrated within-NSFG temporal enrichment for multiple reproductive-health survey endpoints. The approach provides a reproducible framework for population-level reproductive-health phenotyping, while future work should pursue independent survey or registry replication, survey-design-aware uncertainty intervals, and cautious applied interpretation.

6 Data availability

The raw data analyzed in this study are publicly available from CDC/NCHS NSFG public-use data portals, including the 2022–2023 public-use files and 2011–2019 public-use releases. Raw individual-level public-use records are not redistributed in this manuscript package. Processed source-data tables used for figures and manuscript tables are included in the `source_data` directory. Public-use data access URLs and parsing notes are documented in the project README and processing scripts.

7 Code availability

Analysis scripts, split definitions, source-data tables, model metadata, and figure assets are included in the reproducibility package submitted with this manuscript. Raw NSFG individual-level public-use records are not redistributed; the scripts point to CDC/NCHS public-use data portals and document the processing workflow. The package is intended to support peer review and can be archived publicly on GitHub or Zenodo if required by journal policy.

8 Ethics approval and consent to participate

This study used de-identified public-use NSFG data released by CDC/NCHS. No new human-subject data were collected, and the analysis involved no direct participant contact or identifiable private information; therefore, separate informed consent and institutional review board approval were not required for this secondary public-use analysis.

9 Competing interests

The authors declare no competing interests.

10 Funding

No external funding was received for this study.

11 Authors' contributions

Feifan Lu contributed to study conceptualization, public-data curation, formal analysis, figure and table generation, and manuscript drafting. Rui Guan supervised the study, reviewed and edited the manuscript, and approved the final version.

12 Supplementary material

Table 5: Development-cycle cluster selection metrics across k=2–8.

k	Silhouette	Davies–Bouldin	Minimum cluster proportion, %	Bootstrap ARI mean	Bootstrap n	Selected
2	0.483	0.847	43.2	0.999	60	False
3	0.515	0.795	4.8	0.996	60	True
4	0.400	1.130	3.9	0.957	60	False
5	0.360	1.056	3.3	0.941	60	False
6	0.340	1.110	2.2	0.880	60	False
7	0.319	1.185	1.6	0.862	60	False
8	0.327	1.162	1.6	0.912	60	False

Table 6: Full temporal-validation phenotype-by-endpoint enrichment with bootstrap intervals. Intervals are stratified cluster percentile bootstrap intervals using public-use VEST/VECL design variables with respondent weights retained.

Endpoint	Phenotype	Events	Weighted prevalence, %	PR (95% CI)	RD, % (95% CI)
Adverse pregnancy-history proxy	P0	47	22.1	4.11 (2.96–5.47)	16.7 (10.5 to 23.9)
Adverse pregnancy-history proxy	P1	220	9.2	1.72 (1.58–1.85)	3.9 (3.0 to 4.7)
Adverse pregnancy-history proxy	P2	7	0.5	0.09 (0.01–0.19)	-4.9 (-5.7 to -4.1)
Contraceptive vulnerability	P0	20	7.4	0.95 (0.57–1.36)	-0.4 (-3.4 to 2.8)
Contraceptive vulnerability	P1	239	10.4	1.33 (1.22–1.45)	2.6 (1.7 to 3.5)
Contraceptive vulnerability	P2	146	5.6	0.72 (0.63–0.81)	-2.2 (-2.9 to -1.4)
Fertility or loss help	P0	44	18.7	1.68 (1.20–2.21)	7.6 (2.2 to 13.4)
Fertility or loss help	P1	397	18.2	1.64 (1.55–1.75)	7.1 (6.0 to 8.3)
Fertility or loss help	P2	114	4.3	0.39 (0.33–0.45)	-6.8 (-7.7 to -5.9)
Impaired fecundity	P0	112	47.0	1.54 (1.31–1.78)	16.6 (9.3 to 23.5)
Impaired fecundity	P1	906	42.7	1.40 (1.35–1.46)	12.3 (10.7 to 13.9)
Impaired fecundity	P2	478	18.4	0.61 (0.55–0.65)	-12.0 (-13.8 to -10.3)
Mistimed/unwanted pregnancy history	P0	177	76.2	2.23 (2.05–2.43)	42.0 (36.5 to 47.5)
Mistimed/unwanted pregnancy history	P1	1,510	67.0	1.96 (1.88–2.05)	32.8 (31.2 to 34.5)
Mistimed/unwanted pregnancy history	P2	51	2.3	0.07 (0.05–0.09)	-31.9 (-34.1 to -29.6)

Table 7: Age-range sensitivity analysis in the 2022–2023 temporal-validation cycle. The phenotype with the highest 15–44 prevalence ratio for each endpoint is shown, then re-evaluated after including females aged 45–49 years. Phenotype assignment used the primary endpoint-excluded encoder without refitting; reassignment agreement for the 15–44 subset was ARI=1.00.

Endpoint	Phenotype	15–44 N	15–44 PR	15–49 N	15–49 PR	Direction
Contraceptive vulnerability	P1	2,228	1.33	2,691	1.26	same enrichment direction
Fertility or loss help	P0	239	1.68	315	1.35	same enrichment direction
Mistimed/unwanted pregnancy history	P0	239	2.23	315	1.98	same enrichment direction
Adverse pregnancy-history proxy	P0	239	4.11	315	3.25	same enrichment direction
Impaired fecundity	P0	239	1.54	315	1.49	same enrichment direction

Table 8: Covariate-adjusted phenotype endpoint enrichment in 2022–2023. Models are survey-weighted one-vs-rest logistic enrichment models adjusted for age, race/ethnicity, education, poverty, insurance, and parity. Confidence intervals are stratified cluster percentile bootstrap intervals using public-use VEST/VECL. These models are descriptive robustness checks and are not causal models.

Endpoint	Phenotype	Adjusted OR	95% CI	Bootstrap n
Adverse pregnancy-history proxy	P0	2.05	1.21–3.63	300
Adverse pregnancy-history proxy	P1	1.66	1.04–2.90	300
Adverse pregnancy-history proxy	P2	0.11	0.01–0.29	300
Contraceptive vulnerability	P0	0.67	0.36–1.01	300
Contraceptive vulnerability	P1	1.85	1.40–2.44	300
Contraceptive vulnerability	P2	0.52	0.38–0.70	300
Fertility or loss help	P0	1.25	0.79–1.88	300
Fertility or loss help	P1	2.50	1.91–3.55	300
Fertility or loss help	P2	0.25	0.17–0.33	300
Impaired fecundity	P0	0.97	0.67–1.32	300
Impaired fecundity	P1	1.14	0.95–1.39	300
Impaired fecundity	P2	0.84	0.68–1.03	300
Mistimed/unwanted pregnancy history	P0	2.23	1.55–3.42	300
Mistimed/unwanted pregnancy history	P1	9.64	7.72–12.92	300
Mistimed/unwanted pregnancy history	P2	0.02	0.01–0.03	300

Table 9: Baseline phenotyping method comparison in the 2022–2023 temporal-validation cycle. Mean maximum prevalence ratio is averaged across prespecified endpoints. Minimum cluster proportion and bootstrap ARI are development/test stability summaries; a high prevalence ratio from a very small cluster should not be interpreted as a stronger method.

Method	Mean max PR	Development bootstrap ARI	Minimum cluster proportion, %	Endpoints summarized
MCA-style one-hot SVD + k-means	1.57	0.774	27.5	5
Raw matrix PCA + k-means	2.26	0.735	0.1	5
SSL embedding + k-means	2.18	0.953	4.9	5
Selected-variable Bernoulli LCA	1.50	0.904	18.5	5

Table 10: Endpoint-excluded versus full-domain encoder sensitivity. Values show the maximum phenotype prevalence ratio for each endpoint in 2022–2023. The primary endpoint-excluded encoder retained endpoint-enrichment patterns despite direct endpoint-variable exclusion.

Endpoint	Endpoint-excluded encoder	Full-domain encoder
Contraceptive vulnerability	1.33	1.29
Fertility or loss help	1.68	1.59
Mistimed/unwanted pregnancy history	2.23	2.00
Adverse pregnancy-history proxy	4.11	2.01
Impaired fecundity	1.54	1.42

Table 11: Subgroup robustness summary. The table lists the highest subgroup-level phenotype endpoint enrichments with at least 50 respondents in the phenotype-subgroup cell. Full subgroup results are provided in source-data tables.

Subgroup	Endpoint	Phenotype	N	Events	PR
parity_group: 0	Mistimed/unwanted pregnancy history	P1	227	186	10.68
parity_group: 0	Adverse pregnancy-history proxy	P1	227	11	9.90
age_group: 15-24	Adverse pregnancy-history proxy	P1	155	16	8.23
age_group: 15-24	Mistimed/unwanted pregnancy history	P1	155	124	7.82
race_ethnicity: 2	Adverse pregnancy-history proxy	P0	83	14	6.72
insurance: 1	Adverse pregnancy-history proxy	P0	101	16	6.06
age_group: 15-24	Fertility or loss help	P1	155	18	5.37
poverty_group: >=200% FPL	Adverse pregnancy-history proxy	P0	97	11	4.01
parity_group: 0	Fertility or loss help	P1	227	51	3.99
age_group: 25-34	Adverse pregnancy-history proxy	P0	84	14	3.58
poverty_group: 100-199% FPL	Adverse pregnancy-history proxy	P0	51	9	3.34
insurance: 2	Adverse pregnancy-history proxy	P0	90	19	2.94

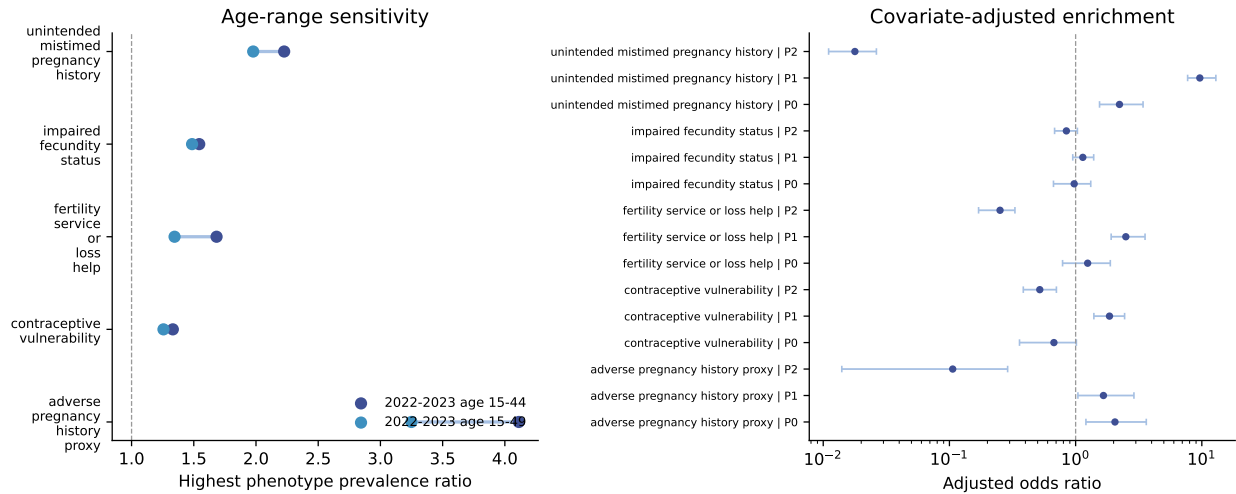


Figure 8: Supplementary robustness analysis for age-range sensitivity and covariate-adjusted endpoint enrichment. The left panel compares highest-phenotype prevalence ratios in 2022–2023 ages 15–44 versus 15–49 years. The right panel shows survey-weighted adjusted odds ratios from one-vs-rest phenotype enrichment models.

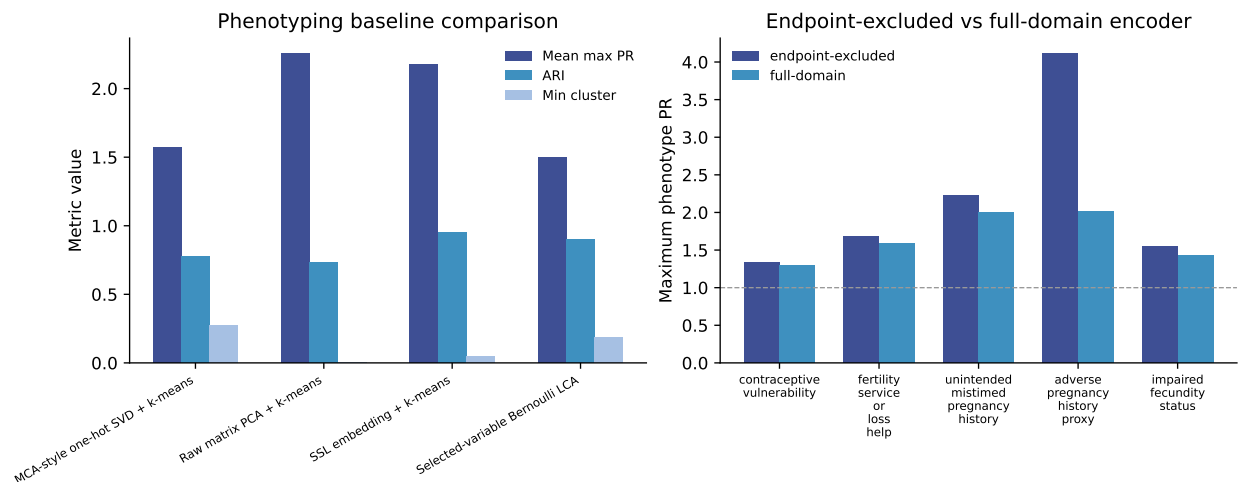


Figure 9: Supplementary baseline-method and leakage-sensitivity analyses. The left panel compares SSL phenotyping with raw PCA, MCA-style one-hot SVD, and selected-variable Bernoulli LCA baselines. The right panel compares maximum endpoint prevalence ratios for the primary endpoint-excluded encoder and a full-domain sensitivity encoder.

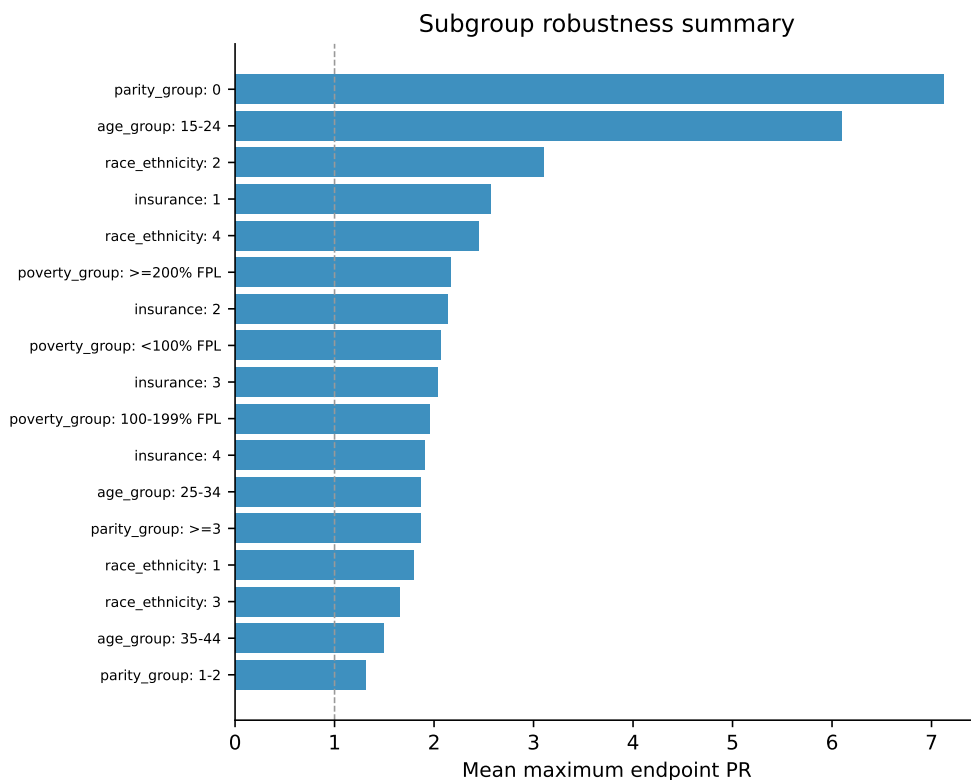


Figure 10: Supplementary subgroup robustness summary across age, race/ethnicity, poverty, insurance, and parity strata. Values summarize mean maximum endpoint prevalence ratios by subgroup; full subgroup endpoint-by-phenotype estimates are included in source data.

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